

$$\begin{cases} \partial_t f = \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \\ f(t=0) = f_0 \end{cases}$$

Lypunov-Schmidt method

- Linearise PDE around $f \equiv 1/2\pi$: operator $\mathcal{L}$

- Compute the kernel of L (i.e. at the bifurcation point)

◦ In our case $\dim(L) \equiv 4$

• Project the PDE onto the basis of the kernel

Lots of integrals, but doable

oGivestheoDE

Antdynamics

Ant dynamics

$$\begin{cases} \partial_t f &= \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \\ f(t=0) &= f_0 \end{cases}$$

Lyapunov-Schmidt method

- Linearise PDE around $f = 1/2\pi$: operator L
- Compute the kernel of L (i.e. at the bifurcation point)
 - In our case $\dim(L) = 4$
- Project the PDE onto the basis of the kernel
 - Lots of integrals, but doable
 - Gives the ODE

Ant dynamics

$$\begin{cases} \partial_t f &= \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta))) f \\ f(t=0) &= f_0 \end{cases}$$

- What happens as $t \rightarrow \infty$?